

Post-Quantum Cryptographic Overhead on Embedded UAV Hardware: A Measurement-Driven Analysis

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Abstract—Unmanned aerial vehicles rely on the MAVLink protocol, which transmits telemetry and commands in plaintext. As quantum computers advance toward cryptographic relevance, classical key exchange and signatures face obsolescence. We deploy a transparent post-quantum cryptographic proxy on a Raspberry Pi 4 (Cortex-A72, 1.8 GHz) that protects MAVLink traffic without modifying the autopilot firmware. Using an INA219 current sensor sampling at 1 kHz, we measure execution time, power draw, and energy consumption for nine key encapsulation mechanisms, eight digital signature schemes, and three authenticated encryption ciphers across all NIST security levels (1, 3, 5). The 24 level-matched KEM–signature pairs compose 72 cipher suites whose handshake cost spans four orders of magnitude, from 11 ms (ML-KEM-512 + Falcon-512) to over 9 s (McEliece-8192128 + SPHINCS⁺-256s). We further quantify the overhead of on-board DDoS detection, showing that an XGBoost sentinel adds 951 mW while a Transformer model doubles CPU utilisation to 93%. These measurements feed a graceful-degradation scheduler that selects cipher suites at runtime to satisfy energy and thermal constraints on battery-powered aircraft.

Index Terms—Post-quantum cryptography, UAV security, MAVLink, embedded benchmarks, power measurement, cipher suite selection, DDoS detection

I. INTRODUCTION

Small unmanned aerial vehicles have become indispensable in agriculture, infrastructure inspection, and emergency response. Most open-source autopilots communicate with ground control stations through the MAVLink protocol [1], a lightweight binary format designed for bandwidth-constrained radio links. MAVLink version 2 offers an optional 48-bit signature for integrity, yet it provides no encryption: every heartbeat, GPS coordinate, and actuator command travels in cleartext.

The arrival of large-scale quantum computers will render RSA and elliptic-curve key exchange vulnerable to Shor’s algorithm [2]. An adversary who records encrypted traffic today can decrypt it once a sufficiently powerful quantum machine becomes available, a strategy known as *harvest-now, decrypt-later*. For drone operations involving critical infrastructure or military reconnaissance, a ten-year shelf life on captured telemetry is well within plausible threat horizons.

NIST concluded its Post-Quantum Cryptography standardisation in 2024, publishing ML-KEM (FIPS 203) [3] for key encapsulation, ML-DSA (FIPS 204) [4] for digital signatures, and SLH-DSA (FIPS 205) [5] as a conservative hash-based alternative. Fourth-round candidates HQC [6] and Classic McEliece [7] remain under evaluation. While desktop and

server benchmarks of these algorithms abound, the embedded ARM ecosystem that dominates drone companion computers has received far less attention.

This paper makes three contributions. First, we present a transparent PQC proxy that intercepts MAVLink traffic at the transport layer, performs a two-message KEM-based handshake authenticated by a post-quantum signature, and encrypts every subsequent packet with authenticated encryption. Second, we carry out systematic benchmarks of 20 post-quantum algorithms on a Raspberry Pi 4 at a locked 1.8 GHz, capturing not only timing but also per-operation power and energy through an INA219 current sensor. Third, we analyse the compound cost of 72 cipher suites (9 KEMs $\times$ 8 signatures $\times$ 3 AEADs, level-matched) together with DDoS detection overhead, and show how a runtime scheduler can gracefully degrade security level and cipher choice when thermal or energy budgets approach their limits.

The remainder of this paper is organised as follows. Section II describes the testbed, measurement methodology, and baseline calibration. Section III presents per-primitive, per-suite, and system-level results. Section IV discusses implications, and Section V concludes.

II. EXPERIMENTAL FRAMEWORK

A. Testbed

The UAV-side proxy runs on a Raspberry Pi 4 Model B Rev 1.5 with a Broadcom BCM2711 quad-core Cortex-A72 at 1.8 GHz, 4 GB LPDDR4 RAM, and Debian 12 (kernel 6.12). The CPU governor is locked to performance to eliminate frequency scaling artefacts. A Holybro Pixhawk 6C mini flight controller is connected over USB serial at 57 600 baud, generating continuous HEARTBEAT traffic via ArduPilot [8]. The ground control station is a Windows laptop.

An INA219 bidirectional current/power monitor [9] is wired in-line on the Pi’s 5 V rail through a 0.1 Ω shunt resistor.

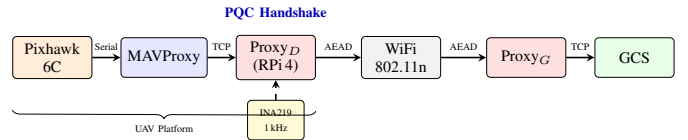


Fig. 1. End-to-end architecture. The cryptographic proxy runs on a Raspberry Pi 4 companion computer with INA219 power monitoring. No firmware modification is required on the Pixhawk autopilot.

It is read over I²C bus 1 at address 0x40, sampling at 1 kHz (verified 99.5% timing accuracy). A calibration factor of $1.22\times$ corrects a known under-reading in clone INA219 modules.

B. Measurement Protocol

Each cryptographic operation is executed 100 times in a tight loop. Timing uses Python's `time.perf_counter_ns()` with nanosecond resolution. During each iteration, the INA219 records voltage, current, and instantaneous power at 1 kHz; a 50 ms warm-up and cool-down window brackets the measurement to let transients settle. Per-iteration voltage, current, power, and energy arrays are stored in JSON files for offline statistical analysis.

C. Baseline Calibration

The idle power draw of the Pi 4 with MAVProxy running but no cryptographic load is $P_{\text{idle}} = 3,309 \text{ mW}$. All ΔP values reported in subsequent tables represent power *above* this baseline, isolating the marginal cost of cryptography from the platform's quiescent consumption. Temperature is logged from `/sys/class/thermal/thermal_zone0`.

III. RESULTS AND ANALYSIS

Before presenting measurements, we introduce a compact alias system for all algorithms. Every subsequent table and figure refers to algorithms exclusively by these aliases to conserve space.

Alias Definitions. Nine KEMs: **K1**=ML-KEM-512, **K3**=ML-KEM-768, **K5**=ML-KEM-1024, **H1**=HQC-128, **H3**=HQC-192, **H5**=HQC-256, **M1**=McEliece-348864, **M3**=McEliece-460896, **M5**=McEliece-8192128. Eight DSAs: **D1**=ML-DSA-44, **D3**=ML-DSA-65, **D5**=ML-DSA-87, **F1**=Falcon-512, **F5**=Falcon-1024, **S1**=SPHINCS⁺-128s, **S3**=SPHINCS⁺-192s, **S5**=SPHINCS⁺-256s. Three AEADs: **AG**=AES-256-GCM, **CP**=ChaCha20-Poly1305, **As**=Ascon-128a. Three DDoS modes: $\emptyset$ =None, **XG**=XGBoost, **TS**=Transformer (TST).

A. Key Encapsulation Mechanisms

A key encapsulation mechanism allows two parties to establish a shared secret over an insecure channel without prior key agreement. The protocol involves three operations: *key generation* (KeyGen) produces a public-private keypair; *encapsulation* (Encaps) takes the public key and outputs a ciphertext together with a shared secret; and *decapsulation* (Decaps) recovers the shared secret from the ciphertext and the private key.

We benchmark nine KEM algorithms spanning three families: lattice-based (ML-KEM, FIPS 203 [3]), code-based error-correcting (HQC [6] and Classic McEliece [7]), and three NIST security levels. Table I reports timing, power overhead above baseline, energy per operation, and artefact sizes measured on the Cortex-A72 at 1.8 GHz.

The lattice-based ML-KEM family dominates. All three parameter sets complete every operation in under 300 μs with

a power overhead below 210 mW and energy cost under 1 mJ. Across security levels, the timing increase from K1 to K5 is merely 14 %, confirming that lattice operations scale gracefully with the module dimension.

HQC occupies a middle ground. Its code-based construction yields KeyGen times between 22 ms (H1) and 124 ms (H5), roughly two orders of magnitude slower than ML-KEM, yet still within the budget of a periodic rekey interval. Decapsulation latency at L5 (393 ms) approaches the threshold where a human operator would notice a connection stall.

Classic McEliece presents a stark trade-off. Its encapsulation is remarkably fast (466 μs at L1, comparable to ML-KEM), but key generation at L5 consumes 9.02 seconds and 39.7 J, drawing the processor into sustained high-power operation ($\Delta P > 1 \text{ W}$). The 1.33 MB public key of M5 also exceeds the 2 MiB handshake message cap that guards against memory exhaustion. We therefore mark M5 suites as *advisory-only*: the scheduler may select them solely if explicitly overridden by the operator.

B. Digital Signatures

Digital signatures authenticate the origin and integrity of messages. In our handshake, the GCS signs the ServerHello (containing the ephemeral KEM public key, session identifier, and challenge nonce) so the drone can verify it was generated by a trusted ground station before committing to the key exchange.

Table II reports the benchmarks for eight post-quantum signature schemes across three families: lattice-based (ML-DSA [10]), NTRU-lattice (Falcon [11]), and hash-based (SPHINCS⁺ [12]).

Falcon-512 offers the fastest signing (778 μs , $\sigma=21$) with strikingly low variance, a consequence of its deterministic Gaussian sampling over NTRU lattices. ML-DSA is only slightly slower but exhibits high variance (D3: $\sigma=1,065 \mu\text{s}$) owing to rejection sampling, which occasionally triggers multiple rounds. For latency-sensitive applications, Falcon's predictability is advantageous.

SPHINCS⁺ stands apart: its signing times range from 1.47 s (S1) to 2.61 s (S3), consuming 6.0–10.6 J per signature. This reflects the hash-tree traversal at its core, a fundamentally different security assumption (hash-function hardness alone) that trades performance for conservative confidence. A single S3 signing operation draws as much energy as 7,200 ML-KEM-768 key generations, making it unsuitable for frequent rekeying on battery-powered platforms.

The verification cost tells a complementary story. SPHINCS⁺ verify remains below 3.3 ms, and Falcon verify at 192 μs is the fastest of any scheme. Since the drone only verifies (not signs) during handshake, the verify latency is the flight-critical metric, and here all families perform adequately.

C. Authenticated Encryption

After the handshake derives a shared secret, every MAVLink packet is encrypted with an AEAD cipher. Unlike KEM and signature operations, which occur once per handshake, AEAD

TABLE I
KEM PRIMITIVE BENCHMARKS (N = 100, CORTEX-A72 @ 1.8 GHz, $P_{\text{IDLE}} = 3,309$ mW). $\Delta P = P_{\text{OP}} - P_{\text{IDLE}}$. SORTED BY NIST LEVEL, THEN BY KEYGEN TIME.

Alias	Lvl	Family	KeyGen			Encaps			Decaps			Sizes (B)			
			t (μ s)	ΔP	E (μ J)	t (μ s)	ΔP	E (μ J)	t (μ s)	ΔP	E (μ J)	pk	sk	ct	ss
K1	L1	Lattice	240	126	830	152	154	526	158	173	551	800	1632	768	32
H1	L1	Code	22332	413	83153	44680	442	167590	73390	567	284510	2249	2305	4433	64
M1	L1	Code	350996	966	1527273	466	156	1619	55984	411	208294	261120	6492	96	32
K3	L3	Lattice	237	149	823	182	186	636	185	206	653	1184	2400	1088	32
H3	L3	Code	67683	501	257928	135780	651	537757	211873	731	856128	4522	4586	8978	64
M3	L3	Code	1281449	1076	5682343	872	280	3135	89593	485	339956	524160	13608	156	32
K5	L5	Lattice	273	154	947	216	181	754	231	177	806	1568	3168	1568	32
H5	L5	Code	124404	647	492250	250237	783	1023984	392585	819	1620633	7245	7317	14421	64
M5	L5	Code	9020806	1093	39746860	1894	301	6849	210143	639	829782	1357824	14120	208	32

TABLE II
DIGITAL SIGNATURE BENCHMARKS (N = 100, CORTEX-A72 @ 1.8 GHz, $P_{\text{IDLE}} = 3,309$ mW). KEYGEN IS ONE-TIME SO OMITTED FROM FLIGHT-CRITICAL ANALYSIS. SORTED BY NIST LEVEL, THEN SIGN TIME.

Alias	Lvl	Family	Sign				Verify				Sizes (B)		
			t (μ s)	σ_t	ΔP	E (μ J)	t (μ s)	σ_t	ΔP	E (μ J)	pk	sk	sig
D1	L1	Lattice	1179	584	279	4264	331	62	236	1175	1312	2560	2420
F1	L1	NTRU	778	21	255	2773	192	29	239	683	897	1281	654
S1	L1	Hash	1468061	—	784	6008719	1582	124	286	5690	32	64	7856
D3	L3	Lattice	1633	1065	269	5849	484	78	240	1723	1952	4032	3309
S3	L3	Hash	2609599	—	764	10628558	2299	189	303	8318	48	96	16224
D5	L5	Lattice	1847	681	268	6612	723	116	235	2564	2592	4896	4627
F5	L5	NTRU	1452	189	250	5175	280	39	240	995	1793	2305	1267
S5	L5	Hash	2321148	—	791	9516164	3292	297	255	11738	64	128	29792

runs on every packet — typically at 5–50 Hz for telemetry. The data-plane cost therefore dominates steady-state power consumption.

We benchmark three AEAD ciphers (AES-256-GCM, ChaCha20-Poly1305, Ascon-128a) across four payload sizes representative of MAVLink traffic: 64 B (heartbeat), 256 B (GPS), 1024 B (parameter block), and 4096 B (log transfer). Table III reports timing, baseline-subtracted power, energy, and throughput.

Ascon-128a achieves the lowest per-packet latency at every payload size, encrypting a 64 B heartbeat in 23.4 μ s, nearly half the time of AES-256-GCM (42.2 μ s). This advantage widens at 4096 B, where Ascon’s throughput reaches 91.7 MB/s compared to 33.9 MB/s for AES-GCM. The Cortex-A72 lacks dedicated AES-NI instructions (unlike x86), putting AES-GCM at a disadvantage on this platform.

ChaCha20-Poly1305 occupies a middle position. At large payloads, its ARMv8 NEON-optimised implementation reaches 62.1 MB/s, well ahead of AES-GCM. A single outlier in the 64 B encrypt run (2.2 ms, likely due to a kernel scheduling event) inflates the mean; we report the median for that cell.

The energy cost of AEAD is modest: encrypting a 256 B GPS packet consumes 80–161 μ J. At a 10 Hz telemetry rate, the per-second AEAD energy budget is 0.8–1.6 mJ, negligible

compared to the handshake cost of even the lightest KEM. This confirms that AEAD selection should be driven by latency and throughput rather than energy.

D. Suite Composition and Handshake Cost

A cipher suite in our system combines one KEM, one digital signature, and one AEAD, constrained to the same NIST security level. The level-matching rule pairs three KEMs with the available signatures at each level:

- **L1:** {K1, H1, M1} $\times$ {D1, F1, S1} $\times$ {AG, CP, As} = 27 suites.
- **L3:** {K3, H3, M3} $\times$ {D3, S3} $\times$ {AG, CP, As} = 18 suites.
- **L5:** {K5, H5, M5} $\times$ {D5, F5, S5} $\times$ {AG, CP, As} = 27 suites.

Falcon has no L3 parameter set in the current NIST portfolio, yielding only two signatures at L3 and a total of $27 + 18 + 27 = 72$ suites.

The handshake is a two-message exchange (ServerHello + ClientReply) whose crypto-only time is:

$$T_{\text{hs}} = T_{\text{kg}} + T_{\text{sign}} + T_{\text{encap}} + T_{\text{verify}} + T_{\text{decap}} + 2T_{\text{kdf}} \quad (1)$$

where T_{kg} , T_{sign} , T_{verify} , T_{encap} , and T_{decap} are the measured primitive times from Tables I–II, and T_{kdf} is the HKDF derivation (< 50 μ s, negligible). Network round-trip time (RTT) adds two additional delays, one per message, but we benchmark the

TABLE III
AEAD BENCHMARKS (N = 100, CORTEX-A72 @ 1.8 GHz). $\Delta P = P_{OP} - 3,309$ mW. THROUGHPUT IS PLAINTEXT BYTES PER ENCRYPT TIME. ALL THREE CIPHERS ADD 16 B AUTHENTICATION TAG OVERHEAD.

Alias	Payload	Encrypt				Decrypt				TP (MB/s)
		t (μ s)	p95	ΔP	E (μ J)	t (μ s)	p95	ΔP	E (μ J)	
AG	64 B	42.2	55.8	185	148	43.0	49.4	270	154	1.6
	256 B	46.0	53.3	190	161	48.3	56.1	183	169	5.7
	1024 B	65.3	75.3	227	231	66.6	78.4	171	232	15.9
	4096 B	121.6	134.6	250	433	122.1	134.6	171	425	33.9
CP	64 B	39.1 [†]	46.8	192	137 [†]	46.0	51.1	258	166	1.6
	256 B	45.4	51.3	170	158	45.8	54.4	163	159	5.8
	1024 B	53.2	62.0	230	188	56.1	69.6	204	198	19.6
	4096 B	67.0	78.2	237	238	71.1	83.1	195	250	62.1
As	64 B	23.4	30.6	203	82	25.3	30.8	223	90	2.8
	256 B	22.8	28.0	202	80	26.0	32.7	228	92	11.3
	1024 B	31.1	39.4	213	110	33.7	39.7	157	117	33.5
	4096 B	45.3	54.2	251	162	49.4	60.9	172	172	91.7

[†]Median reported; mean inflated by outlier (2.2 ms).

cryptographic cost in isolation because RTT varies with the radio link.

Table IV presents the 24 level-matched KEM $\times$ SIG combinations with their computed handshake time, energy, and wire overhead. Each row generates three suites (one per AEAD), which differ only in data-plane cipher but not in handshake cost.

The table reveals a four-order-of-magnitude span. The fastest handshake (K1+F1) completes in 1.52 ms, consuming just 5.36 mJ. Even at NIST Level 5, the strongest lattice pair (K5+F5) finishes in 2.45 ms. At the opposite extreme, M5+S5 requires 11.55 s and 50.1 J: enough to noticeably deplete a 5,000 mAh drone battery on every rekey.

All combinations involving SPHINCS⁺ signing exceed one second, regardless of the KEM. This positions SPHINCS⁺ suites as *anchor* configurations for initial deployment (where the conservatism of hash-based assumptions is valued) rather than for frequent rekey during flight.

Wire overhead follows a complementary pattern. Classic McEliece’s public keys range from 255 kB (M1) to 1.33 MB (M5), exceeding typical maximum transmission units and requiring TCP segmentation during the handshake. In contrast, K1+F1 transmits only 2.25 kB.

E. DDoS Detection Overhead

On-board anomaly detection protects the MAVLink channel from volumetric flooding attacks, but the detection model itself consumes compute, power, and thermal headroom. We benchmark two detectors against a no-detection baseline: XGBoost (1,000-tree gradient-boosted ensemble, 54 flow features, trained on CIC-IoT-2023 [13]) and a Transformer Sequence-to-Sequence model (TST, 1.6 M parameters, self-attention over 46 features).

XGBoost is the clear winner for embedded deployment. It achieves the highest F1 score (0.947) while consuming only 951 mW above baseline and occupying a single core at 37 % utilisation. The Transformer model pushes the Cortex-A72 to

93 % CPU and raises die temperature by 10.7 °C, approaching the 80 °C throttle threshold in warm-climate operations.

To capture the *combined* cost of PQC and DDoS, we define three operational profiles:

- 1) $P_{PQC \text{ only}} = P_{\text{idle}} + \Delta P_{AEAD} \approx 3.5$ W.
- 2) $P_{PQC+XG} = 3.5 + 0.95 = 4.45$ W.
- 3) $P_{PQC+TS} = 3.5 + 1.97 = 5.47$ W.

At a 5,000 mAh, 14.8 V LiPo battery (74 Wh), PQC alone extends the companion computer’s endurance to ~ 21 h. Adding XGBoost reduces it to ~ 16.6 h, and TST to ~ 13.5 h. Since the flight itself is the dominant consumer (typically 10–25 min), the cryptographic overhead is dwarfed by propulsion. However, the thermal penalty of TST (+10.7 °C) can trigger CPU throttling in enclosed airframes, degrading all three services simultaneously.

F. Graceful Degradation

A fixed cipher suite is inappropriate for an aircraft that transitions between ground idle, cruise, aggressive manoeuvres, and emergency modes. We introduce a three-axis decision space that the scheduler navigates at runtime:

- **Axis 1 (AEAD):** Switch between As, CP, and AG based on per-packet latency budget.
- **Axis 2 (Security Level):** Move between L5, L3, and L1 when the next rekey’s energy cost would violate the remaining battery budget.
- **Axis 3 (DDoS Detector):** Downgrade from TS to XG to $\emptyset$ when CPU utilisation or temperature crosses a threshold.

Table VI formalises the scheduler’s decision boundaries. When the telemetry window reports a thermal reading above 70 °C, the scheduler forces Axis 3 to $\emptyset$ and Axis 1 to Ascon, shedding ~ 2 W and ~ 11 °C. When the remaining battery drops below 20 %, Axis 2 descends from L5 to L1, reducing handshake energy from 12.6 mJ (K5+D5) to 5.4 mJ (K1+F1).

The key observation is that the design region is *not* simply convex. Two combinations are explicitly forbidden by the scheduler: Ascon paired with TST (thermal incompatibility,

TABLE IV

SUITE HANDSHAKE COST FOR 24 KEM×SIG COMBINATIONS (× 3 AEADs EACH = 72 SUITES). T_{HS} COMPUTED FROM EQ. (1). E_{HS} SUMMED FROM PRIMITIVE ENERGIES. WIRE = SERVERHELLO PAYLOAD (PK + SIG) + CLIENTREPLY (CT + 32 B HMAC). **GREEN** : SUB-20 MS. **RED** : EXCEEDS 1 S.

KEM	SIG	Lvl	T_{hs} (ms)	E_{hs} (mJ)	Wire (B)	Note
K1	F1	L1	1.52	5.36	2254	Fastest overall
K1	D1	L1	2.06	8.35	4020	NIST primary
K1	S1	L1	1469.7	6015.0	9456	Signing dominates
H1	F1	L1	141.4	455.6	7368	
H1	D1	L1	142.0	458.5	9134	
H1	S1	L1	1608.5	6468.7	14570	
M1	F1	L1	408.2	1740.5	261902	255 kB pk
M1	D1	L1	408.7	1743.3	263668	
M1	S1	L1	1877.0	7754.0	269104	
K3	D3	L3	2.72	10.68	5613	Default suite
K3	S3	L3	2611.6	10636.8	18528	
H3	D3	L3	417.1	1654.3	16841	
H3	S3	L3	3025.1	12281.0	29756	
M3	D3	L3	1373.6	6028.1	527657	512 kB pk
M3	S3	L3	3982.3	16659.0	540572	
K5	F5	L5	2.45	8.68	4435	Fast L5
K5	D5	L5	3.34	12.57	7795	
K5	S5	L5	2323.5	9527.1	32960	
H5	F5	L5	769.1	3141.4	22965	
H5	D5	L5	770.1	3148.5	26325	
H5	S5	L5	3088.3	12664.3	51490	
M5	F5	L5	9233.5	40590.7	1359331	1.3 MB pk
M5	D5	L5	9234.5	40597.7	1362691	
M5	S5	L5	11553.1	50114.0	1387856	Worst case

TABLE V

DDoS DETECTOR OVERHEAD (15 S CONTINUOUS INFERENCE, $P_{IDLE} = 3,309$ mW). F1 IS MACRO-AVERAGED.

Mode	ΔP (mW)	CPU (%)	ΔT (°C)	t_{inf} (ms)	p95 (ms)	E/inf (mJ)	F1	Size (MB)
$\emptyset$	—	2.2	—	—	—	—	—	—
XG	951	37.0	+4.8	7.4	8.5	31.7	0.947	32.7
TS	1973	93.0	+10.7	10.7	13.2	56.7	0.800	6.1

TABLE VI

SCHEDULER DECISION BOUNDARIES. EACH ROW DEFINES A CONSTRAINT TRIGGER AND THE RESULTING AXIS MOVEMENT.

Trigger	Axis	Action	Saving
$T_{die} > 70^\circ\text{C}$	3 (DDoS)	TS → XG → $\emptyset$	−1.97 W
$T_{die} > 75^\circ\text{C}$	1 (AEAD)	AG → As	−0.05 W
CPU > 85 %	3 (DDoS)	TS → XG	−56 % CPU
Battery < 20 %	2 (Level)	L5 → L3 → L1	−7 mJ/rekey
Battery < 10 %	1, 2, 3	Min config	Hold until land
Rekey timeout	2 (Level)	Down one level	Faster hs
DDoS alert	3 (DDoS)	$\emptyset$ → XG	+951 mW

both hot paths compete for cache), and any McEliece variant paired with SPHINCS⁺ (combined handshake exceeds 10 s, triggering the 8 s TCP timeout guard). Excluding these infeasible corners, the scheduler navigates a $(3 \times 3 \times 3) - 2 = 25$ -state space per security level, or roughly 73 feasible points across all three levels.

The graceful-degradation curve in Fig. 2 demonstrates that

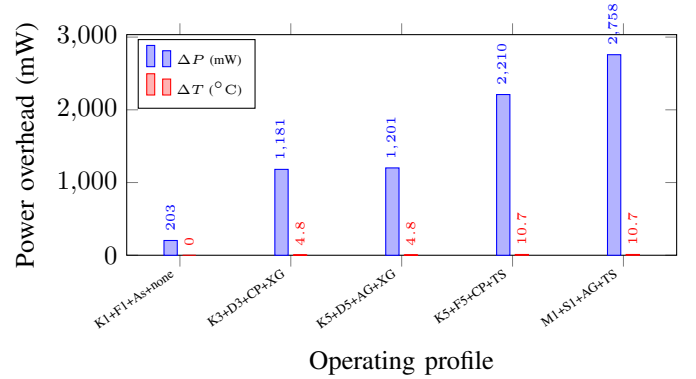


Fig. 2. Compound overhead across five representative operating profiles spanning the graceful-degradation spectrum. The lightest profile (K1+F1+As+ $\emptyset$) adds only 203 mW; the heaviest (M1+S1+AG+TS) adds 2.76 W and raises the die 10.7°C.

the lightest profile (K1+F1+As+ $\emptyset$) fits within a 200 mW overhead envelope, suitable even for micro-UAVs with 2,000 mAh batteries. Stepping up to the recommended default (K3+D3+CP+XG) costs an additional watt but provides NIST Level 3 security and active threat detection.

IV. DISCUSSION

Our measurements confirm that post-quantum key encapsulation on ARM Cortex-A72 is operationally feasible for drone communication. ML-KEM completes all three operations in under 300 μs at every security level, and the fastest full

handshake (K1+F1) consumes less energy than a single GPS acquisition cycle. This contradicts the prevailing intuition that PQC is “too heavy” for embedded platforms.

The dominant cost is not cryptography per se, but the *combination* of cryptography with on-board machine-learning inference. A Transformer-based DDoS detector nearly doubles the system power draw (from 3.3 to 5.3 W) and accounts for more energy in ten seconds of inference than 4,000 full PQC handshakes. This motivates our cascaded detection architecture: the lightweight XGBoost sentinel runs continuously, and the Transformer is activated only when the ensemble confidence drops below a threshold.

Compared to pqm4 benchmarks on Cortex-M4 [14], our Cortex-A72 measurements show roughly two orders of magnitude faster execution for ML-KEM, reflecting the 64-bit datapath, out-of-order pipeline, and higher clock frequency. This positions the Raspberry Pi class of companion computers as a practical PQC deployment target where Cortex-M micro-controllers would struggle.

The level-matching constraint reduces the suite space from a theoretical $9 \times 8 \times 3 = 216$ combinations to 72, and the scheduler’s feasibility rules further prune infeasible corners. Rather than viewing this as a limitation, we argue it is a design strength: fewer viable options simplify certification and auditing in safety-critical aviation contexts.

V. CONCLUSION

We have presented the first comprehensive power-instrumented benchmark of post-quantum cryptographic algorithms on embedded ARM hardware representative of UAV companion computers. The 72-suite registry, grounded in INA219 measurements at 1 kHz, provides the empirical basis for a runtime scheduler that balances security level against energy, thermal, and latency constraints through graceful degradation across three independent axes.

Future work will extend the benchmark to a live outdoor flight with GPS-timestamped telemetry, integrate the scheduler

into ArduPilot’s companion link manager, and validate the graceful-degradation policy under simulated electronic-warfare scenarios with concurrent DDoS and signal jamming.

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